Engineering Mathematics
For IIITH PGEE 2026
By GATE And Tech

Contents

1 Linear Algebra Questions (1–50) **2**

2 Calculus Questions (51–100) **11**

Answers and Explanations **20**

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1 Linear Algebra Questions (1–50)

1. Let A be an $m \times n$ matrix of rank r. Given that $A\mathbf{x} = [1, 0, 0]^T$ has no solution and $A\mathbf{x} = [0, 1, 0]^T$ has exactly one solution. Which of the following must be true?

   (a) $m = 3,\ n = r,\ r < 3$
   (b) $m = 3,\ n > r,\ r = 3$
   (c) $m = 3,\ n = r,\ r = 3$
   (d) $m = 3,\ n > r,\ r < 3$

2. Let A be an $m \times n$ matrix such that $A\mathbf{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has no solution and $A\mathbf{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ has exactly one solution. Consider the following statements:

   (a) $\det(A^T A) = \det(AA^T)$
   (b) $A^T A$ is invertible
   (c) AA^T is positive definite

   Which of the statements must be true?

   (a) Only (2)
   (b) (2) and (3)
   (c) (1) and (2)
   (d) None

3. Let $A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & 1 & 4 \\ -2 & 1 & -4 \end{bmatrix}$. Given that 1 is an eigenvalue of A, which statement is correct?

   (a) The minimal polynomial of A is $(X - 1)(X + 4)$
   (b) The minimal polynomial of A is $(X - 1)^2(X + 4)$
   (c) A is diagonalizable
   (d) $A^{-1} = \frac{1}{4}(M + 3I)$

4. Let A be an $n \times n$ matrix with real entries such that $A^3 = A$. Which of the following is/are necessarily true?

   (a) A is diagonalizable over $\mathbb{R}$
   (b) A is invertible
   (c) $\mathrm{tr}(A)$ is an integer
   (d) All eigenvalues are 0 or 1

5. Let A be a 3×3 matrix with eigenvalues $1, -1, 2$. Then the trace of $\mathrm{adj}(A)$ (the adjugate matrix) is:

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(a) -2
(b) -1
(c) 2
(d) 0

6. Let A be a 3×3 matrix with entries from $\mathbb{R}$ such that $A^2 = 0$. Which of the following is necessarily true?

   (a) $\text{rank}(A) \leq 1$
   (b) A is diagonalizable
   (c) $\text{tr}(A) \neq 0$
   (d) A has a non-zero eigenvalue

7. Let A and B be $n \times n$ matrices such that $AB = A$ and $BA = B$. Which of the following is always true?

   (a) A and B are idempotent
   (b) A and B are invertible
   (c) $A = B$
   (d) $A^2 = B^2 = I$

8. Let V be the vector space of all 2×2 real matrices. Define $T : V \to V$ by $T(X) = AX - XA$, where $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$. Then the dimension of the kernel of T is:

   (a) 1
   (b) 2
   (c) 3
   (d) 4

9. Let A be a 5×5 real matrix with characteristic polynomial $(\lambda - 2)^3(\lambda + 1)^2$. Which of the following is/are possible?

   (a) $\text{rank}(A - 2I) = 2$ and $\text{rank}(A + I) = 4$
   (b) $\text{rank}(A - 2I) = 3$ and $\text{rank}(A + I) = 3$
   (c) $\text{rank}(A - 2I) = 2$ and $\text{rank}(A + I) = 3$
   (d) $\text{rank}(A - 2I) = 1$ and $\text{rank}(A + I) = 4$

10. Let A be an $n \times n$ real symmetric matrix with eigenvalues $\lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$. Then for any non-zero vector $\mathbf{x}$, the Rayleigh quotient $R(\mathbf{x}) = \dfrac{\mathbf{x}^T A \mathbf{x}}{\mathbf{x}^T \mathbf{x}}$ satisfies:

    (a) $\lambda_1 \leq R(\mathbf{x}) \leq \lambda_n$
    (b) $\lambda_1 < R(\mathbf{x}) < \lambda_n$ for all $\mathbf{x}$
    (c) $R(\mathbf{x})$ can be any real number
    (d) $R(\mathbf{x})$ is constant for all $\mathbf{x}$

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11. Let A be a 3×3 orthogonal matrix with determinant 1. Which of the following is necessarily true?

    (a) 1 is an eigenvalue of A
    (b) -1 is an eigenvalue of A
    (c) A is diagonalizable over $\mathbb{R}$
    (d) $A^3 = I$

12. Let A be a 3×3 real matrix such that $A^2 = -I$. Then:

    (a) A has no real eigenvalues
    (b) A is orthogonal
    (c) A is symmetric
    (d) $A^3 = A$

13. Let A be an $n \times n$ matrix with rank r. Then the rank of $\text{adj}(A)$ (the adjugate matrix) is:

    (a) n if $r = n$
    (b) 1 if $r = n - 1$
    (c) 0 if $r < n - 1$
    (d) All of the above

14. Let A be a 5×5 real matrix with characteristic polynomial $\lambda^5 + 2\lambda^4 + 3\lambda^3$. Which of the following is necessarily true?

    (a) A is singular
    (b) A is nilpotent
    (c) $\text{tr}(A) = 0$
    (d) A is diagonalizable

15. Let A be a 3×3 real matrix such that $A^T = -A$ and $A \neq 0$. Then:

    (a) A is singular
    (b) A is invertible
    (c) All eigenvalues are purely imaginary
    (d) A is diagonalizable over $\mathbb{R}$

16. Let A be an $n \times n$ real matrix such that $A^2 = I$ and $A \neq \pm I$. Then:

    (a) A is diagonalizable with eigenvalues ± 1
    (b) A has eigenvalue 1 with multiplicity at least 1
    (c) A has eigenvalue -1 with multiplicity at least 1
    (d) All of the above

17. Let $V = \mathbb{R}^2$ with inner product $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 - x_2 y_1 - x_1 y_2 + 4 x_2 y_2$. Then the condition for vectors $(a, 1)$ and $(1, 2)$ to be orthogonal is:

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(a) $a = 0$
(b) $a = 1$
(c) $a = 2$
(d) $a = 7$

18. The dimension of the subspace of symmetric $n \times n$ real matrices is:

(a) $n(n-1)/2$
(b) $n(n+1)/2$
(c) n^2
(d) n

19. The dimension of the subspace of skew-symmetric $n \times n$ real matrices is:

(a) $n(n-1)/2$
(b) $n(n+1)/2$
(c) n^2
(d) n

20. Let V be the vector space of all real-valued continuous functions on $[-1, 1]$. Let W_e be the subspace of even functions and W_o the subspace of odd functions. Then:

(a) $V = W_e \oplus W_o$
(b) $\dim W_e = \dim W_o = \infty$
(c) Every function can be uniquely written as sum of even and odd functions
(d) All of the above

21. Let $T : \mathbb{R}^4 \to \mathbb{R}^4$ be defined by $T(x_1, x_2, x_3, x_4) = (x_1 + x_2, x_2 + x_3, x_3 + x_4, x_4 + x_1)$. Let $\alpha = (1, 0, 1, 0)$. Define $Z(\alpha, T) = \{f(T)\alpha : f(x) \in \mathbb{R}[x]\}$. Then the dimension of $Z(\alpha, T)$ is:

(a) 1
(b) 2
(c) 3
(d) 4

22. Let $T : V \to V$ be a linear transformation on a finite-dimensional vector space. If $T^2 = T$, then:

(a) $V = \ker T \oplus \operatorname{im} T$
(b) T is diagonalizable
(c) Eigenvalues are 0 and 1 only
(d) All of the above

23. Let $T : V \to V$ be a linear transformation such that $T^2 = -I$. Then:

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(a) T is invertible

(b) T has no real eigenvalues

(c) The dimension of V is even

(d) All of the above

24. Let $T : V \to W$ be a linear transformation with $\dim V = n$, $\dim W = m$. Then $\text{rank}(T) + \text{nullity}(T) = ?$

    (a) n

    (b) m

    (c) $\min(m, n)$

    (d) $\max(m, n)$

25. Let $T : V \to V$ be a linear transformation with minimal polynomial $(X-1)(X-2)$. Then:

    (a) T is diagonalizable

    (b) T has eigenvalues 1 and 2

    (c) $V = \ker(T - I) \oplus \ker(T - 2I)$

    (d) All of the above

26. Let $T : V \to V$ be a linear transformation with characteristic polynomial $(X - 1)^3(X - 2)^2$. Then the possible Jordan forms for T include:

    (a) One Jordan block for eigenvalue 1 of size 3 and one Jordan block for eigenvalue 2 of size 2

    (b) Three Jordan blocks for eigenvalue 1 of sizes $1, 1, 1$ and two blocks for eigenvalue 2 of sizes $1, 1$

    (c) A Jordan block for eigenvalue 1 of size 2 and a block of size 1, and a Jordan block for eigenvalue 2 of size 2

    (d) All of the above are possible depending on the minimal polynomial

27. Let A be a 4×4 real matrix with eigenvalues $1, 1, 2, 3$. If A is diagonalizable, then the minimal polynomial is:

    (a) $(X - 1)(X - 2)(X - 3)$

    (b) $(X - 1)^2(X - 2)(X - 3)$

    (c) $(X - 1)(X - 2)(X - 3)^2$

    (d) $(X - 1)^2(X - 2)(X - 3)^2$

28. Let A be a 3×3 real matrix with characteristic polynomial $\lambda^3 - 3\lambda^2 + 3\lambda - 1$. Then:

    (a) A is nilpotent

    (b) A is invertible

    (c) $A = I$

    (d) A is diagonalizable

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29. Let A be a 4×4 real symmetric matrix with eigenvalues $1, 2, 3, 4$. Then which of the following is true about the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$?

    (a) $Q(\mathbf{x}) > 0$ for all non-zero $\mathbf{x}$
    (b) $Q(\mathbf{x})$ can be negative for some $\mathbf{x}$
    (c) $Q(\mathbf{x}) = 0$ for some non-zero $\mathbf{x}$
    (d) $Q(\mathbf{x})$ is constant on spheres

30. Let A be a 3×3 real symmetric matrix with eigenvalues $1, 2, 2$. Then the number of linearly independent eigenvectors of A is:

    (a) 1
    (b) 2
    (c) 3
    (d) cannot be determined

31. Let A be a 2×2 matrix with $\det(A) = 2$. Then $\det(2A^{-1} + \operatorname{adj}(A))$ is:

    (a) 0
    (b) 2
    (c) 4
    (d) 8

32. Let A be a 3×3 matrix with $\det(A) = 3$. Then $\det(\operatorname{adj}(\operatorname{adj}(A)))$ is:

    (a) 3
    (b) 9
    (c) 27
    (d) 81

33. Let A and B be $n \times n$ matrices such that $AB = I$. Then:

    (a) $BA = I$
    (b) A and B are invertible
    (c) A and B are square matrices of the same size
    (d) All of the above are true

34. Let A be an $n \times n$ matrix such that $A^2 = A$. Then which of the following is false?

    (a) $I - A$ is also idempotent
    (b) A is diagonalizable
    (c) $\operatorname{tr}(A) = \operatorname{rank}(A)$
    (d) A is invertible unless $A = I$

35. Let A be an $n \times n$ matrix with real entries such that $A^T A = I$. Then:

    (a) $\det A = \pm 1$

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(b) All eigenvalues have modulus 1

(c) A is invertible

(d) All of the above

36. Let A be a 3×3 real matrix with eigenvalues $1, 2, 3$. Then the eigenvalues of $\text{adj}(A)$ are:

(a) $6, 3, 2$

(b) $1, 2, 3$

(c) $1/6, 1/3, 1/2$

(d) $6, 6, 6$

37. Let A be a 3×3 real matrix with eigenvalues $1, -1, 0$. Then the rank of A is:

(a) 0

(b) 1

(c) 2

(d) 3

38. Let A be a 4×4 real matrix with eigenvalues $2, 2, 2, 2$. Which of the following is possible?

(a) A is diagonalizable and $A = 2I$

(b) A is not diagonalizable

(c) A is nilpotent

(d) A has rank 1

39. Let A be an $n \times n$ matrix such that $A^3 = A$. Then which of the following is necessarily true?

(a) A is diagonalizable over $\mathbb{C}$

(b) A is invertible

(c) All eigenvalues are 0 or 1

(d) $A^2 = I$

40. Let A be a 3×3 real matrix such that $A^T A = AA^T$. Then A is called normal. Which of the following is true for normal matrices?

(a) A is diagonalizable by a unitary matrix

(b) Eigenvectors corresponding to distinct eigenvalues are orthogonal

(c) A has an orthonormal basis of eigenvectors

(d) All of the above

41. Let A be an $n \times n$ real symmetric positive definite matrix. Then which of the following is true?

(a) All principal minors are positive

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(b) A has a unique positive definite square root

(c) A is invertible

(d) All of the above

42. Let A be a 3×3 real matrix with characteristic polynomial $\lambda^3 - 2\lambda^2 + \lambda$. Then the determinant of A is:

(a) 0

(b) 1

(c) 2

(d) 3

43. Let A and B be $n \times n$ matrices such that AB is invertible. Then:

(a) A and B are both invertible

(b) Either A or B is invertible

(c) A is invertible but B may not be

(d) B is invertible but A may not be

44. Let A be a 3×3 matrix such that $A^2 = 0$ and $A \neq 0$. Then the Jordan canonical form of A is:

(a) $J_1(0) \oplus J_1(0) \oplus J_1(0)$

(b) $J_2(0) \oplus J_1(0)$

(c) $J_3(0)$

(d) Cannot be determined

45. Let A be a 4×4 real matrix with eigenvalues $1, 1, 1, 1$. Which of the following is possible for the Jordan form?

(a) $J_1(1) \oplus J_1(1) \oplus J_1(1) \oplus J_1(1)$

(b) $J_2(1) \oplus J_1(1) \oplus J_1(1)$

(c) $J_3(1) \oplus J_1(1)$

(d) All of the above

46. Let A be a 3×3 real matrix with characteristic polynomial $(\lambda - 1)^3$. Then the minimal polynomial can be:

(a) $(\lambda - 1)$

(b) $(\lambda - 1)^2$

(c) $(\lambda - 1)^3$

(d) All of the above are possible

47. Let A be a 3×3 real matrix such that $A^3 = I$ and $A \neq I$. Then:

(a) A is diagonalizable over $\mathbb{C}$

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(b) Eigenvalues are cube roots of unity

(c) $\mathrm{tr}(A)$ is an integer

(d) All of the above

48. Let A be an $n \times n$ matrix with rank 1. Then:

    (a) A can be written as $\mathbf{u}\mathbf{v}^T$ for some vectors $\mathbf{u}, \mathbf{v}$

    (b) $\mathrm{tr}(A)$ equals the only non-zero eigenvalue

    (c) $A^2 = (\mathrm{tr}\, A)A$

    (d) All of the above

49. Let X be an infinite-dimensional Banach space. Let S be a subset of X such that the linear span of S is all of X. Then:

    (a) S must be countable

    (b) S must be uncountable

    (c) S can be finite

    (d) S must have the same cardinality as X

50. Let P be a finite-dimensional subspace of a Banach space X. Then:

    (a) There exists a bounded linear projection onto P

    (b) There may not exist a bounded projection onto P

    (c) Every projection onto P is bounded

    (d) No projection onto P exists

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2 Calculus Questions (51–100)

51. Suppose $f : \mathbb{R} \to \mathbb{R}$ is differentiable at 0, and sequences $a_n, b_n \to 0$ with $a_n < 0 < b_n$ for all n. Then:

    (a) $\dfrac{f(b_n) - f(a_n)}{b_n - a_n} \to f'(0)$

    (b) The limit exists but may not equal $f'(0)$

    (c) The limit may not exist

    (d) The limit equals $f'(0)$ only if f is continuously differentiable

52. If f is differentiable in a neighborhood of 0 and f' is continuous at 0, then for any sequences $a_n, b_n \to 0$ with $b_n > a_n$:

    (a) $\dfrac{f(b_n) - f(a_n)}{b_n - a_n} \to f'(0)$

    (b) The limit exists but may not equal $f'(0)$

    (c) The limit may not exist

    (d) The limit equals $f'(0)$ only if a_n and b_n are on opposite sides of 0

53. Give an example of a function f such that $\dfrac{f(b_n) - f(a_n)}{b_n - a_n} \to f'(0)$ for all sequences $a_n < b_n \to 0$, but f is not differentiable at points arbitrarily close to 0.

    (a) $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$

    (b) $f(x) = x$ for rational x, 0 for irrational x

    (c) $f(x) = |x|$

    (d) $f(x) = x^3$

54. Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function. Which of the following is true?

    (a) f is continuous everywhere

    (b) f is differentiable everywhere

    (c) f is differentiable except at a countable set of points

    (d) Both A and C

55. Let $f : \mathbb{R} \to \mathbb{R}$ be a convex function. Then:

    (a) f is differentiable almost everywhere

    (b) f has a right derivative and left derivative at every point

    (c) The set of points where f is not differentiable is countable

    (d) All of the above

56. For a strictly convex function f, Jensen's inequality states $f\left(\int u\right) \leq \int f(u)$. Equality holds iff:

    (a) u is constant almost everywhere

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(b) u is linear

(c) u is measurable

(d) u is bounded

57. Suppose $f : (0, \infty) \to \mathbb{R}$ is continuous and $\lim_{n\to\infty} f(n^2 x) = a$ for every $x > 0$. Then:

(a) $\lim_{x\to\infty} f(x) = a$

(b) $\lim_{x\to\infty} f(x)$ may not exist

(c) f is constant

(d) f is periodic

58. Let $f : [0, 1] \to \mathbb{R}$ be a bounded Lebesgue measurable function. Suppose for every x there is a function g_x such that $f = g_x$ a.e. and $\lim_{t\to x} g_x(t)$ exists. Then:

(a) There is a continuous function g such that $g = f$ a.e.

(b) f is continuous a.e.

(c) f is Riemann integrable

(d) All of the above

59. The limit $\lim_{(x,y)\to(0,0)} \dfrac{x^2 y}{x^4 + y^2}$:

(a) Exists and equals 0

(b) Exists and equals 1

(c) Does not exist

(d) Exists and equals ∞

60. The function $f(x, y) = \dfrac{x^3 y}{x^6 + y^2}$ for $(x, y) \neq (0, 0)$, $f(0, 0) = 0$ is:

(a) Continuous at $(0, 0)$

(b) Not continuous at $(0, 0)$

(c) Differentiable at $(0, 0)$

(d) Both A and C

61. Let $f(x, y) = x^2 y + xy^2$. Then the directional derivative of f at $(1, 1)$ in the direction of the vector $(1, 2)$ is:

(a) $\dfrac{3}{\sqrt{5}}$

(b) $\dfrac{6}{\sqrt{5}}$

(c) $\dfrac{9}{\sqrt{5}}$

(d) $\dfrac{12}{\sqrt{5}}$

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62. The equation $x^2 + y^2 + z^2 = 4$ defines a surface. The tangent plane at $(1, 1, \sqrt{2})$ is:

(a) $x + y + \sqrt{2}\, z = 4$
(b) $x + y + \sqrt{2}\, z = 2$
(c) $x + y + \sqrt{2}\, z = 0$
(d) $x + y + \sqrt{2}\, z = 1$

63. Let $f(x, y) = e^{xy}$. Then the second order Taylor expansion about $(0, 0)$ is:

(a) $1 + xy + \dfrac{x^2y^2}{2}$
(b) $1 + xy$
(c) $1 + x + y + xy$
(d) $1 + x + y + xy + \dfrac{x^2}{2} + \dfrac{y^2}{2}$

64. The critical points of $f(x, y) = x^3 - 3xy + y^3$ are:

(a) $(0, 0)$ only
(b) $(1, 1)$ only
(c) $(0, 0)$ and $(1, 1)$
(d) $(0, 0),\ (1, 1),\ (-1, -1)$

65. For $f(x, y) = x^3 - 3xy + y^3$, the point $(1, 1)$ is a:

(a) Local minimum
(b) Local maximum
(c) Saddle point
(d) Not a critical point

66. For $f(x, y) = x^3 - 3xy + y^3$, the point $(0, 0)$ is a:

(a) Local minimum
(b) Local maximum
(c) Saddle point
(d) Not a critical point

67. The solution to the differential equation $\dfrac{dy}{dx} = \dfrac{x + y}{x},\ y(1) = 0$ is:

(a) $y = x \ln x$
(b) $y = x \ln x - x$
(c) $y = x \ln x + x$
(d) $y = \ln x$

68. The differential equation $(3a^2x^2 + by \cos x)\, dx + (2 \sin x - 4ay^2)\, dy = 0$ is exact if:

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(a) $a = 3,\ b = 2$
(b) $a = 2,\ b = 3$
(c) $a = 3,\ b = 4$
(d) $a = 2,\ b = 5$

69. If $2 - Px + Qx^2 = 0$, then a particular integral is:

(a) $y = x^2$
(b) $y = 1/x$
(c) $y = e^x$
(d) $y = e^{-x}$

70. Consider the differential equation $x^2y'' - 2xy' + 2y = 6$ with $y(0) = 3,\ y'(0) = 1$. Then:

(a) $y(x) = cx^2 + x + 3$ is not a solution
(b) $y(x) = cx^2 + x + 3$ is a unique solution
(c) $y(x) = cx^2 + x + 3$ is a solution but not unique
(d) None of these

71. The double integral $\iint_D (x+y)\, dA$ where D is the triangle with vertices $(0,0),\ (1,0),\ (0,1)$ equals:

(a) $\frac{1}{3}$
(b) $\frac{1}{2}$
(c) $\frac{2}{3}$
(d) 1

72. The volume under the plane $z = x + y$ over the rectangle $[0,1] \times [0,2]$ is:

(a) 1
(b) 2
(c) 3
(d) 4

73. The iterated integral $\int_0^1 \int_{\sqrt{x}}^1 e^{y^3}\, dy\, dx$ equals:

(a) $\frac{e-1}{3}$
(b) $\frac{e-1}{2}$
(c) $e - 1$

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(d) $2(e-1)$

74. The line integral $\int_C (x\,dy - y\,dx)$ where C is the circle $x^2+y^2=1$ traversed counterclockwise equals:

    (a) 0
    (b) π
    (c) 2π
    (d) 4π

75. The flux of $\mathbf{F} = (x, y, z)$ through the unit sphere oriented outward is:

    (a) 0
    (b) π
    (c) 2π
    (d) 4π

76. Let $f : \mathbb{R} \to \mathbb{R}$ be an L^1 function. Then f and its Fourier transform cannot both have compact support unless:

    (a) $f = 0$ a.e.
    (b) f is continuous
    (c) f is infinitely differentiable
    (d) f is periodic

77. If f and g are in $L^2(\mathbb{T})$, then $f * g$ is:

    (a) Continuous
    (b) Differentiable
    (c) In L^2
    (d) In L^1

78. Let $F(t) = \chi_{[0,t]}$ for $t > 0$, as a map from $(0, \infty)$ to $L^2(\mathbb{R})$. Then:

    (a) F is differentiable everywhere
    (b) F is nowhere differentiable
    (c) F is differentiable almost everywhere
    (d) F is continuous but not differentiable

79. Define $T : L^2[0,1] \to L^2[0,1]$ by $(Tf)(x) = \int_0^x f(t)\,dt$. Then:

    (a) T is compact
    (b) T has eigenvalues
    (c) T is invertible
    (d) T is self-adjoint

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80. For the operator T in Q79, the spectrum of T is:

(a) $\{0\}$
(b) $\{\lambda : |\lambda| \leq 1\}$
(c) $\{\lambda : |\lambda| = 1\}$
(d) $\mathbb{C}$

81. Suppose $f : \mathbb{R} \to \mathbb{R}$ is 2π-periodic and $f \in L^2(-\pi, \pi)$. If $\sum (|\hat{f}(n)|\, n^k)^2 < \infty$ for some integer $k \geq 1$, then:

(a) f is equal a.e. to a $C^{(k-1)}$ function
(b) f is C^k
(c) f is analytic
(d) f is continuous

82. The Wronskian $W(y_1, y_2, y_3)$ of three solutions to a third order linear homogeneous ODE:

(a) Is identically zero
(b) Is constant
(c) Is either identically zero or never zero
(d) Can be any function

83. Consider the differential equation $(D^3 + 3D^2 + 2D)y = x^2 + 4x + 8$. The particular integral is:

(a) A polynomial of degree 3 with integer coefficients
(b) A polynomial of degree 2 with integer coefficients
(c) A polynomial of degree 3 with rational coefficients
(d) A polynomial of degree 4 with real coefficients

84. The critical point $(0, 0)$ for the system $x' = x + 3y,\ y' = 3x + 2y$ is:

(a) Stable saddle point
(b) Unstable saddle point
(c) Unstable node
(d) Stable node

85. The solution of the differential equation $\left(x\dfrac{dy}{dx} - y\right)^2 = x^2\left(\dfrac{dy}{dx}\right)^2$ is:

(a) $\log(y/x) - 1/x = c$
(b) $\log x - x/y = c$
(c) $\log(x/y) - 1/x = c$
(d) $\log x + x/y = c$

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86. The solution to the initial value problem $y'' + 4y = 0,\ y(0) = 1,\ y'(0) = 0$ is:

(a) $y = \cos 2x$
(b) $y = \sin 2x$
(c) $y = \cosh 2x$
(d) $y = e^{2x}$

87. The solution to $y'' - 4y' + 4y = 0$ with $y(0) = 1,\ y'(0) = 2$ is:

(a) $y = e^{2x}$
(b) $y = e^{2x} + xe^{2x}$
(c) $y = e^{2x} + 2xe^{2x}$
(d) $y = e^{2x} + 3xe^{2x}$

88. The method of undetermined coefficients for $y'' + y = \sin x$ suggests a particular solution of the form:

(a) $A\cos x + B\sin x$
(b) $Ax\cos x + Bx\sin x$
(c) $Ax\cos x$
(d) Ae^{ix}

89. The power series solution to $y' = y$ about $x = 0$ is:

(a) $\sum \frac{x^n}{n!}$
(b) $\sum x^n$
(c) $\sum nx^n$
(d) $\sum (-1)^n \frac{x^n}{n!}$

90. The radius of convergence of the power series $\sum (n!)\, x^n$ is:

(a) 0
(b) 1
(c) ∞
(d) e

91. Let $f : [0, 1] \to \mathbb{R}$ be a continuous function such that $\lim_{n\to\infty} f(n^2 x) = a$ for every x. Then:

(a) $\lim_{x\to\infty} f(x) = a$
(b) f is constant
(c) f is periodic
(d) f is bounded

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92. Let $f : [0,1] \to \mathbb{R}$ be a Lebesgue measurable function. The image $f([0,1])$:

(a) Is always Lebesgue measurable
(b) Need not be Lebesgue measurable
(c) Is always Borel measurable
(d) Is always an interval

93. There exists a function g equal to f a.e. such that the image under g of any closed subset of $[0,1]$ is an F_σ set. This is:

(a) Always possible
(b) Possible only if f is continuous
(c) Possible only if f is Borel measurable
(d) Never possible

94. Let E be a compact subset of $C([0,1])$ with the sup norm. Then:

(a) E is equicontinuous
(b) E is closed and bounded
(c) E is sequentially compact
(d) All of the above

95. For $f \in L^1(\mathbb{R})$, $\left|\int f(x)\,dx\right| \leq \int |f(x)|\,dx$. If $\int |f| = 1$ and $\hat{f}(\xi) = \int e^{-ix\xi} f(x)\,dx$, then:

(a) $|\hat{f}(\xi)| < 1$ for all but possibly one ξ
(b) $|\hat{f}(\xi)| \leq 1$ always
(c) $|\hat{f}(\xi)| = 1$ for all ξ
(d) $|\hat{f}(\xi)|$ can be > 1

96. Let $f(x) = \dfrac{1}{1+x^2}$. Then its Fourier transform is:

(a) $\pi e^{-|\xi|}$
(b) $\sqrt{\pi/2}\, e^{-|\xi|}$
(c) $\pi e^{-\xi^2}$
(d) $\sqrt{\pi}\, e^{-\xi^2/4}$

97. The series $\sum \dfrac{1}{n^2}$ converges to:

(a) $\dfrac{\pi^2}{6}$
(b) $\dfrac{\pi^2}{8}$
(c) $\dfrac{\pi^2}{4}$

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(d) $\frac{\pi^2}{2}$

98. The series $\sum \frac{(-1)^n}{n}$ converges to:

    (a) $-\ln 2$
    (b) $\ln 2$
    (c) $\frac{\pi}{4}$
    (d) 1

99. Let f be a continuous function on $[0,1]$ such that $\int_0^1 f(x)x^n\,dx = 0$ for all $n = 0, 1, 2, \ldots$. Then:

    (a) $f \equiv 0$
    (b) f may be non-zero
    (c) f is odd
    (d) f is constant

100. Let $f : \mathbb{R} \to \mathbb{R}$ be uniformly continuous and integrable. Then:

    (a) $\lim_{|x|\to\infty} f(x) = 0$
    (b) f is bounded
    (c) f is square integrable
    (d) All of the above

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Answers and Explanations

1. **Correct Answer: A**
   Explanation: From $A\mathbf{x} = [1, 0, 0]^T$ having no solution, the system is inconsistent, implying the rank r is less than the number of equations, so $r < m = 3$. From $A\mathbf{x} = [0, 1, 0]^T$ having exactly one solution, the null space is trivial, meaning the columns are linearly independent, so $r = n$. Therefore, $m = 3,\ n = r,\ r < 3$.

2. **Correct Answer: A**
   Explanation: As in Q1, the conditions imply A has full column rank ($r = n$) and $r < m = 3$. Therefore, $A^T A$ ($n \times n$) is invertible, so statement (2) is true. AA^T is a 3×3 singular matrix (rank $n < 3$), so its determinant is zero and it cannot be positive definite, making (1) and (3) false.

3. **Correct Answer: B**
   Explanation: The characteristic polynomial is calculated as $(\lambda - 1)^2(\lambda + 4)$. The eigenvalue 1 has algebraic multiplicity 2. Solving $(A - I)x = 0$ yields a one-dimensional eigenspace (geometric multiplicity 1), so the matrix is not diagonalizable and has a Jordan block of size 2 for eigenvalue 1. Therefore, the minimal polynomial is $(X - 1)^2(X + 4)$.

4. **Correct Answer: A and C**
   Explanation: $A^3 = A \Rightarrow A^3 - A = 0 \Rightarrow A(A - I)(A + I) = 0$. The minimal polynomial divides $X(X - 1)(X + 1)$, which has distinct linear factors, so A is diagonalizable over $\mathbb{R}$ (A is true). The eigenvalues are from $\{0, 1, -1\}$, so the trace, being the sum of eigenvalues, is an integer (C is true). A is not necessarily invertible (eigenvalue 0 possible), and eigenvalues can be -1 as well, so B and D are false.

5. **Correct Answer: B**
   Explanation: For a 3×3 matrix, the eigenvalues of $\text{adj}(A)$ are the products of the other two eigenvalues: $(-1) \times 2 = -2,\ 1 \times 2 = 2,\ 1 \times (-1) = -1$. Their sum (the trace) is $-2 + 2 - 1 = -1$.

6. **Correct Answer: A**
   Explanation: $A^2 = 0$ means A is nilpotent. Its only eigenvalue is 0, so the trace is 0 and it has no non-zero eigenvalue. A non-zero nilpotent matrix is not diagonalizable. A nilpotent matrix of index 2 on a 3-dimensional space must have Jordan form $J_2(0) \oplus J_1(0)$, which has rank 1. Thus $\text{rank}(A) \leq 1$.

7. **Correct Answer: A**
   Explanation: From $AB = A$, multiply on the right by A: $ABA = A^2$. Using $BA = B$, we get $A(BA) = AB = A$. So $A^2 = A$. Similarly, $B^2 = B$. Hence A and B are idempotent. They need not be equal, invertible, or square to identity.

8. **Correct Answer: B**
   Explanation: The kernel consists of matrices that commute with A. A is a Jordan block with eigenvalue 1. Matrices that commute with such a block are polynomials in A, which form a vector space of dimension 2 (spanned by I and $A - I$).

9. **Correct Answer: A, B, C are possible; D is impossible.**
   Explanation: For eigenvalue 2 (algebraic multiplicity 3), geometric multiplicity

(gm) can be 1, 2, or 3. $\text{rank}(A - 2I) = 5 - \text{gm}$, so possible ranks are 4, 3, 2. For eigenvalue -1 (algebraic multiplicity 2), gm can be 1 or 2, so $\text{rank}(A+I) = 5-\text{gm}$, giving possible ranks 4 or 3. Thus possible pairs $(\text{rank}(A - 2I), \text{rank}(A + I))$ are $(2, 4), (3, 3), (2, 3), (3, 4), (4, 4)$. Option D $(1, 4)$ requires gm $= 4$ for eigenvalue 2, which is impossible as gm cannot exceed algebraic multiplicity (3).

10. **Correct Answer: A**
Explanation: This is the Rayleigh quotient theorem. For a real symmetric matrix, the Rayleigh quotient $R(\mathbf{x})$ lies between the smallest and largest eigenvalues, with equality achieved when $\mathbf{x}$ is the corresponding eigenvector.

11. **Correct Answer: A**
Explanation: A 3×3 orthogonal matrix with determinant 1 represents a rotation. In 3D, every rotation has an axis of rotation, which corresponds to an eigenvector with eigenvalue 1. It may not be diagonalizable over $\mathbb{R}$ if the other eigenvalues are complex, and $A^3 = I$ is not necessarily true.

12. **Correct Answer: A**
Explanation: If $A^2 = -I$, then for any eigenvalue λ and eigenvector v, $A^2v = \lambda^2 v = -v$, so $\lambda^2 = -1$, hence $\lambda = \pm i$. Thus A has no real eigenvalues. Orthogonality or symmetry is not implied.

13. **Correct Answer: D**
Explanation: For an $n \times n$ matrix A:

    - If $\text{rank}(A) = n$, then A is invertible and $\text{adj}(A) = \det(A)\, A^{-1}$, so $\text{rank}(\text{adj}(A)) = n$.
    - If $\text{rank}(A) = n - 1$, then $\text{adj}(A) \neq 0$ and $\text{rank}(\text{adj}(A)) = 1$.
    - If $\text{rank}(A) < n - 1$, then all $(n - 1) \times (n - 1)$ minors of A are zero. Hence $\text{adj}(A) = 0$, and therefore $\text{rank}(\text{adj}(A)) = 0$.

    Thus, all the statements are true.

14. **Correct Answer: A**
Explanation: The characteristic polynomial is $\lambda^5 + 2\lambda^4 + 3\lambda^3 = \lambda^3(\lambda^2 + 2\lambda + 3)$. Since $\lambda = 0$ is an eigenvalue with multiplicity at least 3, $\det A = 0$, so A is singular. Trace = sum of roots = -2 (from the coefficient of λ^4), not zero. It is not nilpotent because of the quadratic factor with non-zero roots, and diagonalizability is not guaranteed.

15. **Correct Answer: A**
Explanation: For a real skew-symmetric matrix of odd order, the determinant is zero. This is because $\det(A) = \det(-A^T) = (-1)^n \det(A) = -\det(A)$ for odd n, implying $\det(A) = 0$. So A is singular. Eigenvalues are either 0 or purely imaginary pairs; 0 must occur. Over $\mathbb{R}$, it is not diagonalizable if non-zero.

16. **Correct Answer: D**
Explanation: $A^2 = I$ implies the minimal polynomial divides $X^2 - 1 = (X - 1)(X + 1)$, which has distinct roots, so A is diagonalizable with eigenvalues ± 1. Since $A \neq \pm I$, both $+1$ and -1 must appear. All statements are true.

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17. **Correct Answer: D**
Explanation: Orthogonality condition: $\langle (a,1),(1,2)\rangle = a\cdot 1 - 1\cdot 1 - a\cdot 2 + 4\cdot 1\cdot 2 = a - 1 - 2a + 8 = -a + 7 = 0 \Rightarrow a = 7$.

18. **Correct Answer: B**
Explanation: Symmetric matrices are determined by entries on and above the diagonal: n diagonal entries $+\, n(n-1)/2$ off-diagonal entries $= n(n+1)/2$.

19. **Correct Answer: A**
Explanation: Skew-symmetric matrices have zeros on the diagonal and the entries below the diagonal are determined by those above. The number of independent entries is the number above the diagonal: $n(n-1)/2$.

20. **Correct Answer: D**
Explanation: Any function f can be uniquely decomposed as $f(x) = \dfrac{f(x)+f(-x)}{2} + \dfrac{f(x)-f(-x)}{2}$, the even and odd parts. Both subspaces are infinite-dimensional. All statements are true.

21. **Correct Answer: B**
Explanation: Compute $T\alpha = (1,1,1,1)$. Then $T^2\alpha = T(1,1,1,1) = (2,2,2,2) = 2(1,1,1,1) = 2T\alpha$. Thus the cyclic subspace is spanned by α and $T\alpha$, which are linearly independent. Hence dimension $= 2$.

22. **Correct Answer: D**
Explanation: $T^2 = T$ implies T is a projection. Its minimal polynomial divides $X^2 - X = X(X-1)$, so eigenvalues are 0 and 1. Minimal polynomial has distinct roots, so T is diagonalizable. Also, $V = \ker T \oplus \operatorname{im} T$. All statements true.

23. **Correct Answer: D**
Explanation: $T^2 = -I$ implies T is invertible with $T^{-1} = -T$. Eigenvalues satisfy $\lambda^2 = -1$, so no real eigenvalues. Taking determinants, $(\det T)^2 = (-1)^{\dim V}$, so $\dim V$ must be even. All statements true.

24. **Correct Answer: A**
Explanation: By the Rank–Nullity Theorem, $\operatorname{rank}(T) + \operatorname{nullity}(T) = \dim(\text{domain}) = n$.

25. **Correct Answer: D**
Explanation: Minimal polynomial has distinct linear factors, so T is diagonalizable. Its eigenvalues are exactly 1 and 2. The vector space decomposes as the direct sum of the eigenspaces. All statements true.

26. **Correct Answer: D**
Explanation: The characteristic polynomial only gives algebraic multiplicities. The Jordan form depends on the minimal polynomial. All listed configurations are possible for some choice of T with that characteristic polynomial.

27. **Correct Answer: A**
Explanation: Diagonalizable implies the minimal polynomial has distinct linear factors. It must be $(X-1)(X-2)(X-3)$, including each distinct eigenvalue only once, regardless of their algebraic multiplicities.

28. **Correct Answer: B**
Explanation: The characteristic polynomial is $(\lambda - 1)^3$. So all eigenvalues are 1, determinant $= 1^3 = 1$, hence A is invertible. A need not be I or diagonalizable (it could be a Jordan block). It is not nilpotent (eigenvalues are 1, not 0).

29. **Correct Answer: A**
Explanation: All eigenvalues are positive, so the matrix is positive definite. Therefore, the quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x} > 0$ for all non-zero $\mathbf{x}$.

30. **Correct Answer: C**
Explanation: A real symmetric matrix is always diagonalizable. Hence geometric multiplicity equals algebraic multiplicity. So eigenvalue 1 has 1 independent eigenvector, eigenvalue 2 has 2 independent eigenvectors, total = 3.

31. **Correct Answer: D**
Explanation: For a 2×2 matrix, $\text{adj}(A) = \det(A)\, A^{-1} = 2A^{-1}$. So $2A^{-1} + \text{adj}(A) = 2A^{-1} + 2A^{-1} = 4A^{-1}$. Then $\det(4A^{-1}) = 4^2 \det(A^{-1}) = 16 \times (1/\det(A)) = 16 \times (1/2) = 8$.

32. **Correct Answer: D**
Explanation: For an $n \times n$ matrix, $\det(\text{adj}(A)) = (\det A)^{n-1}$. Here $n = 3$, so $\det(\text{adj}(A)) = 3^2 = 9$. Then $\det(\text{adj}(\text{adj}(A))) = (\det(\text{adj}(A)))^{n-1} = 9^2 = 81$.

33. **Correct Answer: D**
Explanation: For square matrices, $AB = I$ implies A and B are invertible and are inverses of each other. Hence $BA = I$ as well. All statements are true.

34. **Correct Answer: D**
Explanation: A is idempotent. $(I - A)$ is also idempotent (true). Idempotent matrices are diagonalizable (true). For an idempotent matrix, $\text{tr}(A) = \text{rank}(A)$ (true). However, A is invertible only if rank $= n$, i.e., $A = I$. If A is a non-identity idempotent (e.g., a projection onto a proper subspace), it is singular. So statement D is false.

35. **Correct Answer: D**
Explanation: $A^T A = I$ defines an orthogonal matrix. Such matrices have determinant ± 1, all eigenvalues have modulus 1, and they are invertible with $A^{-1} = A^T$. All statements are true.

36. **Correct Answer: A**
Explanation: $\text{adj}(A) = \det(A)\, A^{-1}$. $\det(A) = 1 \cdot 2 \cdot 3 = 6$. Eigenvalues of A^{-1} are $1, 1/2, 1/3$. So eigenvalues of $\text{adj}(A)$ are $6 \times 1 = 6,\ 6 \times 1/2 = 3,\ 6 \times 1/3 = 2$.

37. **Correct Answer: C**
Explanation: Since 0 is an eigenvalue, $\det A = 0$, so rank < 3. The non-zero eigenvalues 1 and -1 indicate that the matrix has at least two non-zero eigenvalues (counting multiplicity), so its rank is at least 2. Therefore rank = 2.

38. **Correct Answer: A and B are possible; C and D are not.**
Explanation: All eigenvalues are 2, so the matrix is of the form $2I + N$, with N nilpotent. It can be diagonalizable (if $N = 0$, so $A = 2I$) or not (if $N \neq 0$, e.g., a

Jordan block). Nilpotent would require eigenvalue 0, so C is impossible. Rank 1 would require nullity = 3, which is not achievable when all eigenvalues are 2, so D is also impossible.

39. **Correct Answer: A**
Explanation: As in Q4, minimal polynomial divides $X(X-1)(X+1)$, which has distinct linear factors, so A is diagonalizable over $\mathbb{C}$. Eigenvalues can be $0, 1, -1$, so A is not necessarily invertible, and $A^2 = I$ is not necessarily true.

40. **Correct Answer: D**
Explanation: The spectral theorem for normal matrices (over $\mathbb{C}$) states they are unitarily diagonalizable, have an orthonormal basis of eigenvectors, and eigenvectors for distinct eigenvalues are orthogonal. All statements are true.

41. **Correct Answer: D**
Explanation: For a real symmetric positive definite matrix: all principal minors are positive (Sylvester's criterion), it has a unique positive definite square root, and its determinant is positive, so it is invertible. All statements are true.

42. **Correct Answer: A**
Explanation: The characteristic polynomial is $\lambda^3 - 2\lambda^2 + \lambda = \lambda(\lambda^2 - 2\lambda + 1) = \lambda(\lambda-1)^2$. The constant term (up to sign) is $\det A$. For a polynomial $\lambda^n + a_{n-1}\lambda^{n-1} + \cdots + a_0$, $\det A = (-1)^n a_0$. Here $a_0 = 0$, so $\det A = 0$.

43. **Correct Answer: A**
Explanation: If AB is invertible, then $\det(AB) = \det A \cdot \det B \neq 0$. Hence both $\det A \neq 0$ and $\det B \neq 0$, so A and B are both invertible.

44. **Correct Answer: B**
Explanation: A is nilpotent of index 2. The largest Jordan block size is 2. For a 3×3 matrix with $A^2 = 0$ and $A \neq 0$, the only possible Jordan form is one block of size 2 and one block of size 1: $J_2(0) \oplus J_1(0)$.

45. **Correct Answer: D**
Explanation: All eigenvalues are 1. Any partition of 4 into block sizes is possible: $1+1+1+1$ (diagonalizable), $2+1+1$, $3+1$, and 4. All are possible Jordan forms.

46. **Correct Answer: D**
Explanation: The characteristic polynomial is $(\lambda - 1)^3$. The minimal polynomial can be $(\lambda - 1)$ if $A = I$, $(\lambda - 1)^2$ if A has a Jordan block of size 2, or $(\lambda - 1)^3$ if A has a Jordan block of size 3. All are possible.

47. **Correct Answer: D**
Explanation: Minimal polynomial divides $X^3 - 1 = (X-1)(X-\omega)(X-\omega^2)$, which has distinct roots, so A is diagonalizable over $\mathbb{C}$. Eigenvalues are cube roots of unity. Since $A \neq I$, not all eigenvalues are 1, so the trace is $1 + \omega + \omega^2 = 0$, which is an integer. All statements are true.

48. **Correct Answer: D**
Explanation: A rank 1 matrix can be written as an outer product $\mathbf{u}\mathbf{v}^T$. Its only non-zero eigenvalue equals the trace (since trace = sum of eigenvalues). Also, $A^2 = (\mathbf{u}\mathbf{v}^T)(\mathbf{u}\mathbf{v}^T) = \mathbf{u}(\mathbf{v}^T\mathbf{u})\mathbf{v}^T = (\mathbf{v}^T\mathbf{u})\mathbf{u}\mathbf{v}^T = (\operatorname{tr} A)\, A$. All statements are true.

49. **Correct Answer: B**
Explanation: In an infinite-dimensional Banach space, any Hamel basis (algebraic basis) must be uncountable. Since S spans X, it contains a Hamel basis, hence S must be uncountable.

50. **Correct Answer: A**
Explanation: In any normed space, a finite-dimensional subspace is complemented. There exists a continuous linear projection onto it. This is a standard result.

51. **Correct Answer: A**
Explanation: Differentiability at 0 implies $\lim_{h\to 0} \frac{f(h)-f(0)}{h} = f'(0)$. The given quotient can be manipulated using this definition, and the condition $a_n < 0 < b_n$ ensures that the limit is indeed $f'(0)$. This is a standard result.

52. **Correct Answer: A**
Explanation: By the Mean Value Theorem, $\frac{f(b_n)-f(a_n)}{b_n-a_n} = f'(c_n)$ for some c_n between a_n and b_n. Since $c_n \to 0$ and f' is continuous at 0, $f'(c_n) \to f'(0)$.

53. **Correct Answer: A**
Explanation: The function $f(x) = x^2 \sin(1/x)$ for $x \neq 0$, $f(0) = 0$ is differentiable at 0 with $f'(0) = 0$, and its derivative $f'(x) = 2x\sin(1/x) - \cos(1/x)$ is not continuous near 0 (hence f is not differentiable in the classical sense at points accumulating to 0 in terms of having a continuous derivative). However, f is actually differentiable everywhere; it is not *continuously* differentiable near 0. This is the best available option.

54. **Correct Answer: D**
Explanation: A convex function on $\mathbb{R}$ is continuous on its domain. It is differentiable except at a countable set of points. So both A and C are true.

55. **Correct Answer: D**
Explanation: Convex functions have right and left derivatives at every point, are differentiable except at a countable set, and hence differentiable almost everywhere. All statements are true.

56. **Correct Answer: A**
Explanation: For a strictly convex function, equality in Jensen's inequality holds if and only if the function u is constant almost everywhere.

57. **Correct Answer: A**
Explanation: This is a known result: if $\lim_{n\to\infty} f(n^2x) = a$ for every $x > 0$, and f is continuous, then $\lim_{x\to\infty} f(x) = a$. The idea is to use the fact that for any large t, we can find n and x such that t is close to n^2x.

58. **Correct Answer: A**
Explanation: The condition implies that f is "approximately continuous" at every point. A theorem states that such a function is equal almost everywhere to a continuous function.

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59. **Correct Answer: C**
Explanation: Approaching along $y = kx^2$: $\dfrac{x^2(kx^2)}{x^4 + k^2x^4} = \dfrac{k}{1+k^2}$, which depends on k. Hence the limit does not exist.

60. **Correct Answer: B**
Explanation: Approaching along $y = x^3$: $\dfrac{x^3 \cdot x^3}{x^6 + x^6} = \dfrac{x^6}{2x^6} = \dfrac{1}{2} \neq 0$. So the function is not continuous at $(0,0)$.

61. **Correct Answer: C**
Explanation: $\nabla f = (2xy + y^2,\ x^2 + 2xy)$. At $(1,1)$: $(3,3)$. The unit vector in the direction of $(1,2)$ is $(1/\sqrt{5},\ 2/\sqrt{5})$. Directional derivative $= 3 \cdot \dfrac{1}{\sqrt{5}} + 3 \cdot \dfrac{2}{\sqrt{5}} = \dfrac{9}{\sqrt{5}}$.

62. **Correct Answer: A**
Explanation: $\nabla F = (2x, 2y, 2z)$. At $(1,1,\sqrt{2})$: $(2,2,2\sqrt{2})$. Tangent plane: $2(x-1) + 2(y-1) + 2\sqrt{2}(z-\sqrt{2}) = 0 \Rightarrow x + y + \sqrt{2}\,z = 1 + 1 + 2 = 4$.

63. **Correct Answer: B**
Explanation: $f(0,0) = 1$. $f_x = ye^{xy} \to 0$, $f_y = xe^{xy} \to 0$. $f_{xx} = y^2e^{xy} \to 0$, $f_{yy} = x^2e^{xy} \to 0$, $f_{xy} = e^{xy} + xye^{xy} \to 1$. So Taylor expansion: $1 + 0 \cdot x + 0 \cdot y + 1 \cdot xy = 1 + xy$.

64. **Correct Answer: C**
Explanation: $f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$. $f_y = -3x + 3y^2 = 0 \Rightarrow -3x + 3x^4 = 0 \Rightarrow 3x(x^3 - 1) = 0 \Rightarrow x = 0$ or $x = 1$. Points: $(0,0)$ and $(1,1)$.

65. **Correct Answer: A**
Explanation: $f_{xx} = 6x = 6$, $f_{yy} = 6y = 6$, $f_{xy} = -3$. At $(1,1)$: $D = f_{xx}f_{yy} - f_{xy}^2 = 36 - 9 = 27 > 0$ and $f_{xx} > 0$, so local minimum.

66. **Correct Answer: C**
Explanation: At $(0,0)$: $f_{xx} = 0$, $f_{yy} = 0$, $f_{xy} = -3$. $D = 0 - 9 = -9 < 0$, so saddle point.

67. **Correct Answer: A**
Explanation: Rewrite as $y' - \dfrac{1}{x}y = 1$. Integrating factor $e^{-\int (1/x)\,dx} = 1/x$. Then $\dfrac{d}{dx}(y/x) = 1/x$. Integrate: $y/x = \ln x + C$. Using $y(1) = 0$ gives $C = 0$. So $y = x \ln x$.

68. **Correct Answer: A**
Explanation: For exactness, $\dfrac{\partial M}{\partial y} = \dfrac{\partial N}{\partial x}$. Here $M = 3a^2x^2 + by\cos x$, $N = 2\sin x - 4ay^2$. $\partial M/\partial y = b\cos x$, $\partial N/\partial x = 2\cos x$. So $b = 2$. The value of a is not constrained. Option A has $b = 2$.

69. **Correct Answer: B**
Explanation: This likely comes from a Cauchy–Euler or similar ODE context. The function $1/x$ is a common solution to such equations.

70. **Correct Answer: C**
Explanation: Homogeneous solution: $c_1x + c_2x^2$. Particular solution: constant 3.

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So general solution $y = c_1x + c_2x^2 + 3$. Initial conditions at $x = 0$: $y(0) = 3$ gives $3 = 3$ (no info), $y'(0) = 1$ gives $c_1 = 1$. So $y = x + c_2x^2 + 3$, which is of the form $cx^2 + x + 3$ with $c = c_2$ arbitrary. So it is a solution but not unique.

71. **Correct Answer: A**
Explanation: $\int_0^1 \int_0^{1-x} (x+y)\, dy\, dx = \int_0^1 \left[x(1-x) + \frac{(1-x)^2}{2}\right] dx = \int_0^1 \left(\frac{1}{2} - \frac{x^2}{2}\right) dx = \left[\frac{x}{2} - \frac{x^3}{6}\right]_0^1 = \frac{1}{2} - \frac{1}{6} = \frac{1}{3}$.

72. **Correct Answer: C**
Explanation: $\int_0^2 \int_0^1 (x+y)\, dx\, dy = \int_0^2 \left(\frac{1}{2} + y\right) dy = \left[\frac{y}{2} + \frac{y^2}{2}\right]_0^2 = 1 + 2 = 3$.

73. **Correct Answer: A**
Explanation: Swap the order of integration. The region is $0 \le y \le 1$, $0 \le x \le y^2$. Then the integral becomes $\int_0^1 \int_0^{y^2} e^{y^3}\, dx\, dy = \int_0^1 y^2 e^{y^3}\, dy = \frac{1}{3}(e-1)$.

74. **Correct Answer: C**
Explanation: By Green's theorem, $\int_C x\, dy - y\, dx = \iint_D (1 - (-1))\, dA = 2 \times \text{Area}(D) = 2 \times \pi = 2\pi$.

75. **Correct Answer: D**
Explanation: By the Divergence Theorem, flux $= \iiint_V \nabla \cdot \mathbf{F}\, dV = \iiint_V 3\, dV = 3 \times \frac{4}{3}\pi = 4\pi$.

76. **Correct Answer: A**
Explanation: This is the uncertainty principle for Fourier transforms: A non-zero function and its Fourier transform cannot both have compact support. The only possibility is $f = 0$ a.e.

77. **Correct Answer: A**
Explanation: The convolution of two L^2 functions on the circle is continuous. This follows from the fact that their Fourier coefficients are in ℓ^2, and the convolution's Fourier coefficients are the product, which is in ℓ^1, implying a continuous function.

78. **Correct Answer: B**
Explanation: The map $t \mapsto \chi_{[0,t]}$ from $(0, \infty)$ to $L^2(\mathbb{R})$ is nowhere differentiable in the strong sense. This is a classic example of a nowhere differentiable Banach-valued function.

79. **Correct Answer: A**
Explanation: T is the Volterra operator. It is a Hilbert–Schmidt operator, hence compact. It has no eigenvalues (except 0). It is not self-adjoint and not invertible.

80. **Correct Answer: A**
Explanation: The Volterra operator is quasinilpotent, meaning its spectrum is $\{0\}$.

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81. **Correct Answer: A**
Explanation: The condition $\sum(|\hat{f}(n)|\, n^k)^2 < \infty$ means f is in the Sobolev space H^k. In 1D, the Sobolev embedding theorem states that $H^k \hookrightarrow C^{(k-1)}$ (continuous functions with $k-1$ continuous derivatives). So f is equal a.e. to a $C^{(k-1)}$ function.

82. **Correct Answer: C**
Explanation: By Abel's formula, the Wronskian of solutions to a linear homogeneous ODE is either identically zero or never zero.

83. **Correct Answer: C**
Explanation: The operator $D^3+3D^2+2D = D(D^2+3D+2) = D(D+1)(D+2)$. Since the right-hand side is a polynomial of degree 2, and the operator has a factor D, we need to integrate. The particular integral will be a polynomial of degree 3 with rational coefficients.

84. **Correct Answer: B**
Explanation: The matrix is $\begin{bmatrix} 1 & 3 \\ 3 & 2 \end{bmatrix}$. Trace $= 3 > 0$, determinant $= 2 - 9 = -7 < 0$. So it is an unstable saddle point.

85. **Correct Answer: B**
Explanation: This is a Clairaut-type equation. The answer $\log x - x/y = c$ is one of the solutions.

86. **Correct Answer: A**
Explanation: Characteristic equation $r^2 + 4 = 0 \Rightarrow r = \pm 2i$. General solution $y = A\cos 2x + B\sin 2x$. $y(0) = 1 \Rightarrow A = 1$. $y'(0) = 2B = 0 \Rightarrow B = 0$. So $y = \cos 2x$.

87. **Correct Answer: A**
Explanation: Characteristic equation $(r-2)^2 = 0$. General solution $y = (A + Bx)e^{2x}$. $y(0) = 1 \Rightarrow A = 1$. $y' = (B + 2A + 2Bx)e^{2x}$, $y'(0) = B + 2 = 2 \Rightarrow B = 0$. So $y = e^{2x}$.

88. **Correct Answer: B**
Explanation: The homogeneous solution includes $\sin x$ and $\cos x$. Since the forcing term $\sin x$ is a solution of the homogeneous equation, the particular solution must be multiplied by x: $Ax\cos x + Bx\sin x$.

89. **Correct Answer: A**
Explanation: $y' = y$ with $y(0) = 1$ has solution $y = e^x = \displaystyle\sum_{n=0}^{\infty} \frac{x^n}{n!}$.

90. **Correct Answer: A**
Explanation: Ratio test: $\left|\dfrac{a_{n+1}}{a_n}\right| = \dfrac{(n+1)!}{n!} = n + 1 \to \infty$. So radius of convergence $R = 0$.

91. **Correct Answer: A**
Explanation: This is the same result as in Q57. For a continuous function, if the limit along the sequence n^2x exists for every x, then the limit at infinity exists and equals that value.

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92. **Correct Answer: B**
Explanation: The image of a measurable set under a measurable function need not be measurable. This is a classic pathology.

93. **Correct Answer: A**
Explanation: This is a theorem based on Luzin's property: every measurable function is equal almost everywhere to a function with this property.

94. **Correct Answer: D**
Explanation: In $C([0,1])$ with the sup norm, compact subsets are exactly those that are closed, bounded, and equicontinuous (Arzelà–Ascoli theorem). Sequential compactness is equivalent to compactness in metric spaces. So all statements are true.

95. **Correct Answer: B**
Explanation: The Fourier transform of an L^1 function is continuous and bounded by $\|f\|_1$. Since $\int |f| = 1$, we have $|\hat{f}(\xi)| \leq 1$ for all ξ. By the Riemann–Lebesgue lemma, $|\hat{f}(\xi)| \to 0$ as $|\xi| \to \infty$, so it cannot be 1 for all but possibly one ξ unless f is a pure exponential, which is not in L^1.

96. **Correct Answer: A**
Explanation: The Fourier transform (depending on the convention) of $\dfrac{1}{1+x^2}$ is $\pi e^{-|\xi|}$. This is a standard result.

97. **Correct Answer: A**
Explanation: This is the Basel problem: $\displaystyle\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$.

98. **Correct Answer: A**
Explanation: The alternating harmonic series $\displaystyle\sum_{n=1}^{\infty} \frac{(-1)^n}{n} = -\ln 2$ (using the series for $\ln(1+x)$ at $x = -1$).

99. **Correct Answer: A**
Explanation: The set of polynomials is dense in $C[0,1]$ by the Weierstrass approximation theorem. If all moments are zero, then $\int_0^1 f(x)\,p(x)\,dx = 0$ for every polynomial p. By continuity and density, $\int_0^1 f(x)^2\,dx = 0$, so $f \equiv 0$.

100. **Correct Answer: D**
Explanation: A uniformly continuous and integrable function on $\mathbb{R}$ must satisfy $\displaystyle\lim_{|x|\to\infty} f(x) = 0$. Uniform continuity plus limit zero at infinity implies it is bounded (continuous on compact intervals and small outside). Square integrability follows because $\int f^2 \leq (\sup|f|) \int |f| < \infty$. All statements are true.

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